# Maulana Abul Kalam Azad University of Technology, WB
## (Formerly known as West Bengal University of Technology)
## Syllabus of BCA
## Effective from academic session 2023-24

**Graduate Attribute:**

Bachelor of Computer Applications (BCA) programs typically aim to equip students with a well-rounded set of skills and knowledge in the field of computer science and applications. The Graduate Attributes (GA), which characterize the information, competencies, values, and abilities that students acquire in order to develop holistically and contribute to society. These characteristics include traits that can be transferred beyond the sphere of study into the national and international realms and are cultivated through curricular, co-curricular, and extra-curricular initiatives.

1. **Technical Competence:** Graduates should have a solid foundation in computer science concepts, programming languages, software development methodologies, and technical tools commonly used in the industry.
2. **Problem-Solving Skills:** Graduates should be capable of analyzing complex problems, designing effective solutions, and implementing these solutions using appropriate programming languages and technologies.
3. **Software Development:** Graduates should be skilled in the entire software development lifecycle, including requirements analysis, design, implementation, testing, deployment, and maintenance of software applications.
4. **Programming Proficiency:** Graduates should have expertise in at least one or more programming languages and be capable of writing efficient, maintainable, and well-documented code.
5. **Database Management:** Graduates should understand database design principles, be familiar with database management systems, and be able to design and manage databases for various applications.
6. **Communication Skills:** Graduates should possess effective communication skills, including the ability to articulate technical concepts clearly to both technical and non-technical stakeholders.
7. **Teamwork and Collaboration:** Graduates should be able to work effectively as part of a team, demonstrating the ability to collaborate, share ideas, and contribute positively to group projects.
8. **Ethical and Professional Behavior:** Graduates should understand the ethical implications of technology and demonstrate professional conduct in their work, respecting intellectual property rights, privacy, and security considerations.
9. **Continuous Learning:** Graduates should recognize the dynamic nature of the field and the need for ongoing learning and skill development to stay up-to-date with emerging technologies and industry trends.
10. **Adaptability:** Graduates should be adaptable to changing technological landscapes and be able to apply their foundational knowledge to new and evolving challenges.
11. **Critical Thinking:** Graduates should be capable of evaluating information critically, making informed decisions, and applying logical reasoning to solve complex problems.
12. **Project Management:** Graduates should have a basic understanding of project management principles, enabling them to manage time, resources, and tasks effectively within software development projects.
13. **User-Centered Design:** Graduates should appreciate the importance of designing software applications with user experience in mind, ensuring usability, accessibility, and user satisfaction.

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### Curriculum Structure

### FIRST SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC101 | Digital Electronics | 3 | 3 | | |
| 2 | | BCAC191 | Digital Electronics Lab | 2 | | 2 | |
| 3 | | BCAC102 | Programming for Problem Solving through C | 3 | 3 | | |
| 4 | | BCAC192 | Programming for Problem Solving Lab | 2 | | 2 | |
| 5 | DSE | MIM101 | Principles of Management | 3 | 3 | | |
| 6 | GE | | Any one from GE baskets A or D | 3 | 3 | | |
| 7 | AECC | AECC101 | English & Professional Communication | 2 | 2 | | |
| 8 | SEC | SEC181 | Life Skills & Personality Development | 2 | | 2 | |
| 9 | VAC | VAC181 A/ B/C | Yoga/ Health & Wellness/ Sports | 2 | | 2 | |
| Total Credit | | | | 22 | | | |

### SECOND SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC201 | Computer Architecture | 3 | 3 | | |
| 2 | | BCAC291 | Computer Architecture Lab | 2 | | 2 | |
| 3 | | BCAC202 | Basics of Web Design Using Html, CSS, Java Script | 3 | 3 | | |
| 4 | | BCAC292 | Basics of Web Design Using Html, CSS, Java Script Lab | 2 | | 2 | |
| 5 | DSE | | Any one from Minor in Management Basket | 3 | 3 | | |
| 6 | GE | | Any one from GE baskets B or E | 3 | 3 | | |
| 7 | AECC | AECC201 | Modern Indian Languages and Literature | 2 | 2 | | |
| 8 | SEC | SEC281 | IT Skills | 2 | | 2 | |
| 9 | VAC | VAC281A/ B/C/D | Critical Thinking/ NSS/ Mental Health/Environmental Studies | 2 | | 2 | |
| Total Credit | | | | 22 | | | |

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### THIRD SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC301 | Python Programming | 3 | 3 | | |
| 2 | | BCAC391 | Python Programming Lab | 2 | | 2 | |
| 3 | | BCAC302 | Data Structure through C | 3 | 3 | | |
| 4 | | BCAC392 | Data Structure Lab | 2 | | 2 | |
| 5 | DSE | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| 6 | GE | | Any one from GE baskets C or F | 3 | 3 | | |
| 7 | AECC | AECC 301 | The Constitution, Human Rights and Law | 2 | 2 | | |
| 8 | SEC | SEC 381 | Understanding basics of Cyber Security | 2 | | 2 | |
| Total Credit | | | | 21 | | | |

### FOURTH SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC401 | Data Base Management System | 3 | 3 | | |
| 2 | | BCAC491 | DBMS Lab | 2 | | 2 | |
| 3 | | BCAC402 | Operating System | 4 | 3 | | 1 |
| 4 | | BCAC403 | Software Engineering | 4 | 3 | | 1 |
| 5 | DSE | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| 6 | | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| 7 | AECC | AECC 401 | Society Culture and Human Behavior | 2 | 2 | | |
| Total Credit | | | | 23 | | | |

### FIFTH SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC501 | PHP WITH MYSQL | 3 | 3 | | |
| 2 | | BCAC591 | PHP WITH MYSQL LAB | 2 | | 2 | |
| 3 | | BCAC502 | Object Oriented Programming with Java | 3 | 3 | | |
| 4 | | BCAC592 | Object Oriented Programming with Java Lab | 2 | | 2 | |
| 5 | DSE | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| 6 | | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| 7 | SEC | SEC581 | Internship | 4 | | 4 | |
| Total Credit | | | | 22 | | | |

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### SIXTH SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC601 | Advance Java With Web Application | 3 | 3 | | |
| 2 | | BCAC691 | Advance Java With Web Application Lab | 2 | | 2 | |
| 3 | | BCAC602 | Unix and Shell Programming | 3 | 3 | | |
| 4 | | BCAC692 | Unix and Shell Programming Lab | 2 | | 2 | |
| 5 | | BCAC603 | Networking | 4 | 3 | | 1 |
| 6 | DSE | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| 7 | | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| **Total Credit** | | | | **22** | | | |

### SEVENTH SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC701A /BCAC701B /BCAC701C | Introduction to Data Science/ Introduction to Machine Learning /Advanced Data Analysis Tools | 3 | 3 | | |
| 2 | | BCAC791A /BCAC791B /BCAC791C | Data Science Lab with Python /Machine Learning Lab / Advanced Data Analysis Tools Lab | 2 | | 2 | |
| 3 | | BCAC702 | Research Methodology | 3 | 3 | | |
| 4 | | BCAC792 | Research Methodology Lab | 2 | | 2 | |
| 5 | | BCAC703 | Cyber Security | 4 | 3 | | 1 |
| 6 | DSE | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| 7 | | | Any one from Minor in Management Basket | 4 | 3 | | 1 |
| **Total Credit** | | | | **22** | | | |

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### EIGHTH SEMESTER

| Sl No. | Subject Type | Subject Code | Subject Name | Credit Point | Credit Distribution | | |
|---|---|---|---|---|---|---|---|
| | | | | | L | P | T |
| 1 | DSC | BCAC801A/ BCAC801B | Cloud Computing / Artificial Intelligence | 3 | 3 | | |
| 2 | | BCAC891A/ BCAC891B | Cloud Computing Lab / Artificial Intelligence Lab | 2 | | 2 | |
| 3 | | BCAC802 | Statistical Analysis with R Programming | 3 | 3 | | |
| 4 | | BCAC892 | Statistical Analysis with R Programming Lab | 2 | | 2 | |
| 5 | SEC | SEC881 | Major / Research Project | 12 | | 12 | |
| **Total Credit** | | | | **22** | | | |